# Incentives and COVID-19 Vaccinations Replication Project

Molly Marino

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## Contents

# 1 Original Study

The paper "Incentives Can Spur COVID-19 Vaccination Uptake" by Heike Klüver, Felix Hartmann, Macartan Humphreys, Ferdinand Geissler, and Johannes Giesecke investigates whether policy incentives can increase people's willingness to receive a COVID-19 vaccine. The study was conducted during the pandemic to address vaccine hesitancy, which threatened efforts to reach high vaccination coverage and control the spread of COVID-19.

The researchers conducted a factorial survey experiment with 20,500 participants in Germany between March 5 and March 25, 2021. Participants were shown hypothetical scenarios in which policies varied across three dimensions: financial incentives, freedoms for vaccinated individuals, and access to vaccination through local doctors. Financial incentives were either 25 euros or 50 euros. Freedoms referred to liberties that vaccinated participants could enjoy that unvaccinated participants could not. Access through local doctors aimed to reduce barriers and increase trust.

The study found that all three policy strategies increased willingness to vaccinate. For the full sample, a financial incentive of 25 euros increased willingness by one percentage point, while 50 euros increased it by 2.2 percentage points. Freedoms for vaccinated participants increased willingness by 2.5 percentage points, and vaccination at local doctors increased willingness by three percentage points. Among participants who were undecided, effects were stronger: financial incentives increased willingness by five percentage points, freedoms increased willingness by six percentage points, and local doctor access increased willingness by five percentage points. When all three incentives were combined, acceptance among undecided participants increased by thirteen percentage points.

The study also found that older participants were more responsive to local doctor access, while younger participants were more responsive to freedoms. This indicates that demographic factors, particularly age, shape how incentives influence vaccination willingness.

# 2 Methods

The study used linear regression models to estimate the effect of each policy on the likelihood of vaccination. Individual fixed effects controlled for unobserved participant characteristics that did not change over time. Participants rated their willingness to vaccinate on a scale from zero to one for each scenario. To explore heterogeneity in responses, a machine learning method called causal forests was used, examining factors such as age, trust, risk preferences, and prior vaccine hesitancy.

# 3 Data

The replication uses the original dataset analyzed in R, organized in three scripts. The first script, `prep_data.R`, prepares and cleans the dataset. The second script, `main_results.R`, estimates baseline regression models. The third script, `het_effects.R`, examines heterogeneous treatment effects across subgroups. These scripts are compiled through the master document `index.Rmd`.

```
rmarkdown::render("index.Rmd")
```

# 4 Original Model

The analysis estimates the change in willingness to vaccinate between different hypothetical scenarios as a function of treatment. Treatments include financial incentives of 25 euros and 50 euros, freedoms for vaccinated participants, and vaccination at local doctors. Individual fixed-effects regressions are estimated using a stacked dataset with indicators for each participant to control for unobserved characteristics.

```
lm_robust(
  outcome_diff ~ negative_diff + financialA_diff + financialB_diff +
      transaction_diff,
  data = df, se_type = "stata") %>%
  tidy

lm_robust(
  outcome ~ negative + financialA + financialB + transaction,
  data = df_stacked, se_type = "stata", fixed_effects = ~ID) %>%
  tidy

our_analysis <- function(dff)
  lm_robust(
    outcome_diff ~ negative*financialA*transaction + negative*
       financialB*transaction,
    data = dff %>%
      mutate(
        negative = negative_diff - mean(negative_diff),
        financialA = financialA_diff - mean(financialA_diff),
        financialB = financialB_diff - mean(financialB_diff),
        transaction = transaction_diff - mean(transaction_diff)
      ),
    se_type = "stata")

statuses <- unique(df$status)

model_list <- lapply(statuses, function(j) our_analysis(df %>% dplyr
   ::filter(status == j)))

names(model_list) <- statuses

model_list$All <- our_analysis(df)

by_status <-
  lapply(model_list, function(model) tidy(model)) %>%
```

```
  bind_rows(.id = "status") %>%
  mutate(status = factor(status, c("All", statuses)))

policy_experiments <- list(
  experiment_1 =
    df_stacked %>%
    mutate(
      Z = negative * financialB * transaction,
      zip = (1-negative) * (1-financialA) * (1-financialB) * (1-
         transaction)
    ) %>%
    dplyr::filter((Z==1) | (zip==1)) %>%
    group_by(Z) %>%
    mutate(N = n()) %>% ungroup() %>%
    group_by(status, Z) %>%
    summarize(n = n()/mean(N), share_pop_vaccinated = n()*mean(
       outcome)/mean(N)),

  experiment_2 =
    df_stacked %>%
    dplyr::filter(financialA == 0 & negative == 1 & transaction ==
       1) %>%
    group_by(financialB) %>%
    mutate(N = n()) %>% ungroup() %>%
    group_by(status, financialB) %>%
    summarize(n = n()/mean(N), share_pop_vaccinated = n()*mean(
       outcome)/mean(N)) %>%
    arrange(financialB) %>% mutate(Z = financialB)

) %>% bind_rows(.id = "experiment") %>%
  mutate(
    Z = (experiment == "experiment_1") * Z +
      (experiment == "experiment_2") * (Z + .5),
    Z = factor(Z, c(0, .5, 1), c("No incentives", "Local doctors +
       Freedoms", "All incentives"))
  ) %>%
  dplyr::filter(!is.na(Z))

fig_2_ABC <- ggarrange(
  lm_plot + theme(plot.margin=unit(c(1,1,1.5,1.2)*.6,"cm")),
  ggarrange(
    age_plot + ylim(-.03, .13) + theme(plot.margin=unit(c
       (1,1,1.5,1.2)*.6,"cm")),
    distance_plot + ylab("") + ylim(-.03, .13) + theme(plot.margin=
       unit(c(1,1,1.5,1.2)*.6,"cm")),
    labels = c("B", "C"),
    ncol = 2, common.legend = TRUE, legend="bottom"
```

```
  ),
  nrow = 2, labels = "A"
)

lm_robust(
  outcome_diff ~ negative_diff + financialA_diff + financialB_diff +
      transaction_diff,
  data = df, se_type = "stata"
)
```

### 4.1 Figure 1: Estimated Effects on Vaccination Probability

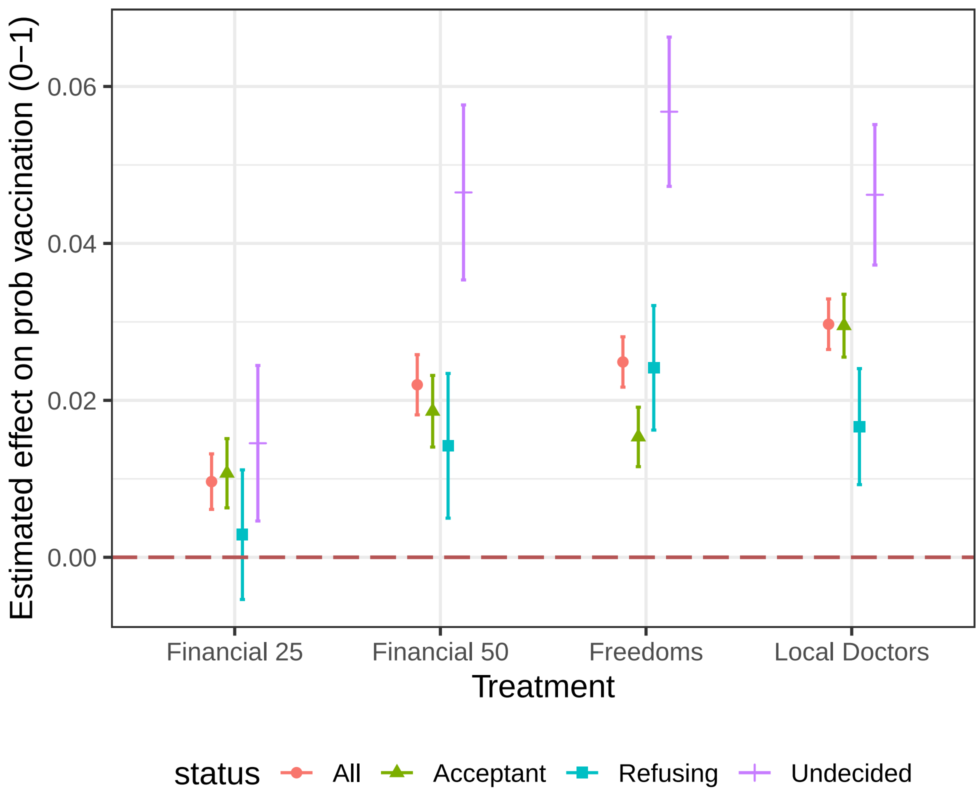


Figure 1: Estimated effect of each incentive on the probability that a participant would take a COVID-19 vaccine. For the overall sample, financial incentives of 25 euros increased willingness by one percentage point, and 50 euros increased it by 2.2 percentage points. Freedoms increased willingness by 2.5 percentage points, and vaccination through local doctors increased willingness by three percentage points. Among undecided participants, financial incentives increased willingness by five percentage points, freedoms by six percentage points, and local doctor access by five percentage points. This figure shows that undecided participants respond most strongly, while those already refusing vaccination show little change.

```
figure_1A <- by_status %>%
  filter(term %in% c("negative", "financialA", "financialB", "
     transaction")) %>%
  filter(status != "Vaccinated") %>%
  mutate(Treatment = factor(term, treatment_levels, treatment_labels
     )) %>%
  ggplot(aes(Treatment, estimate, color = status, shape = status)) +
  geom_point(position = position_dodge(width = 0.3)) +
  geom_errorbar(aes(ymin = conf.low, ymax = conf.high),
                position = position_dodge(width = 0.3), width = 0.1)
                    +
  geom_hline(yintercept = 0, linetype = "longdash", size = 0.75,
     colour = "#B55555") +
  theme_bw(base_size = 14) +
  ylab("Estimated␣effect␣on␣prob␣vaccination␣(0-1)") +
  theme(legend.position = "bottom")
```

## 4.2 Figure 2: Predicted Share of Population Vaccinated

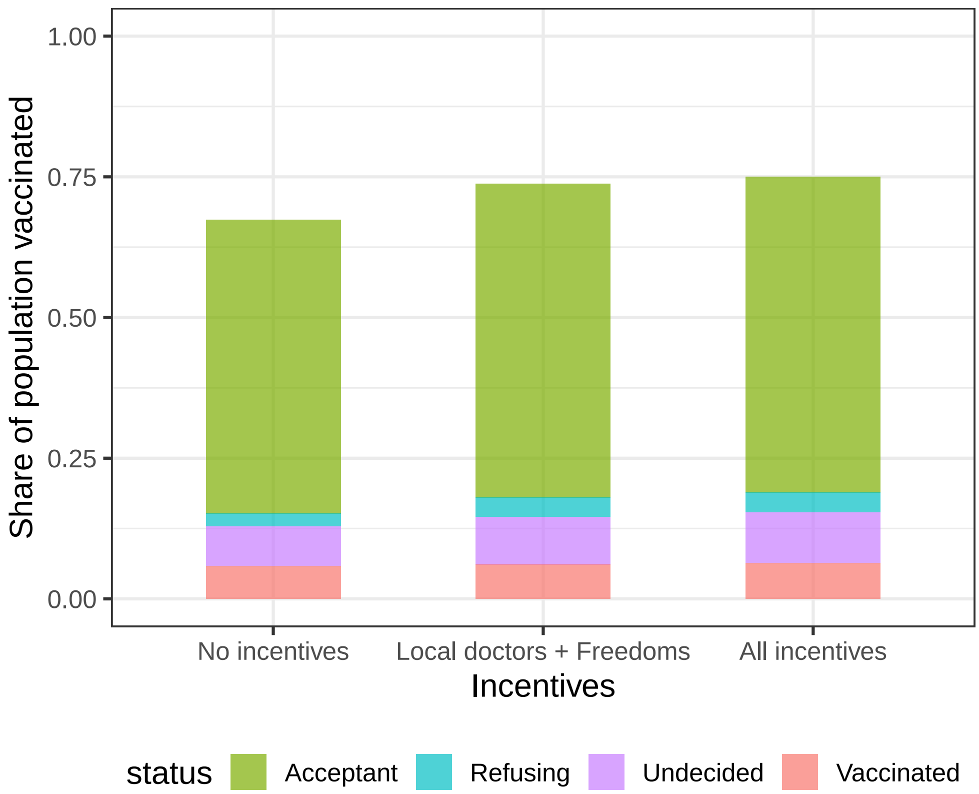


Figure 2: Predicted share of the population that would be vaccinated under different incentive scenarios. In the baseline scenario with no incentives, about 67 percent of participants are predicted to be vaccinated. Introducing freedoms and vaccination at local doctors raises this share to about 73 percent. When all three incentives, including the 50-euro financial incentive, are combined, the vaccinated share rises to approximately 75 percent. These gains are largely due to undecided participants deciding to vaccinate, rather than participants who would refuse vaccination.

```
figure_1B <- policy_experiments %>%
  ggplot(aes(fill = status, y = share_pop_vaccinated, x = Z)) +
  geom_bar(position = "stack", stat = "identity", width = 0.5, alpha
     = 0.7) +
  scale_fill_manual(values = group.colors) +
  ylab("Share␣of␣population␣vaccinated") +
  xlab("Incentives") +
  theme_bw(base_size = 14) +
  ylim(0:1) +
  theme(legend.position = "bottom")
# Figure 3
fig_2_ABC <- ggarrange(
  lm_plot + theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
  ggarrange(
    age_plot + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    distance_plot + ylab("") + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    labels = c("B", "C"),
    ncol = 2,
    common.legend = TRUE,
    legend = "bottom"
  ),
  nrow = 2,
  labels = "A"
)
```

### 4.3 Figure 3: Heterogeneous Treatment Effects

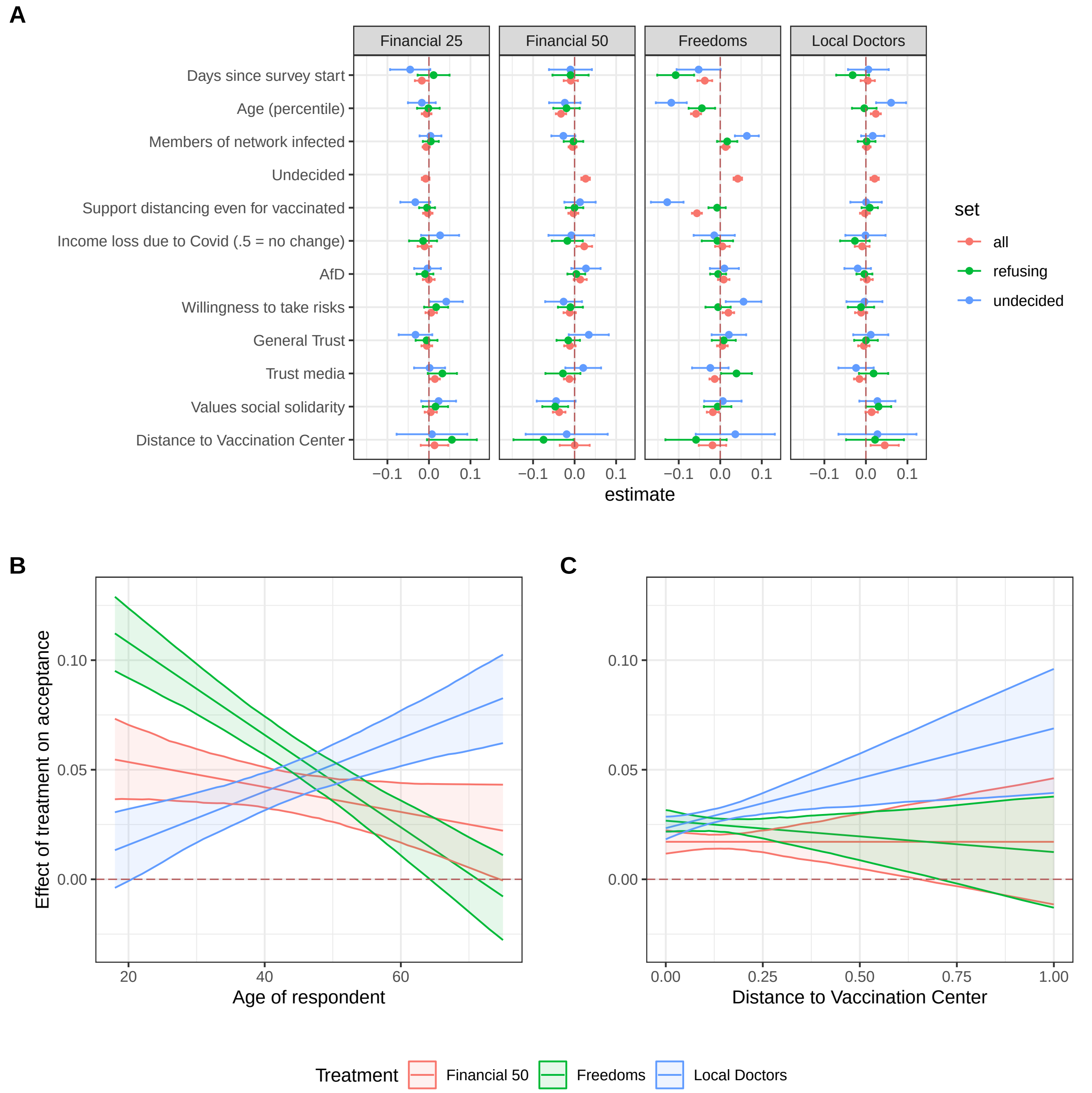


Figure 3: Heterogeneous treatment effects using a causal forest model. Older participants respond more to the opportunity to be vaccinated at local doctors, while younger participants respond more to freedoms. The figure illustrates how demographic and attitudinal factors influence the effectiveness of incentives.

```
fig_2_ABC <- ggarrange(
  lm_plot + theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
  ggarrange(
    age_plot + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    distance_plot + ylab("") + ylim(-0.03, 0.13) +
      theme(plot.margin = unit(c(1,1,1.5,1.2) * 0.6, "cm")),
    labels = c("B", "C"),
    ncol = 2,
    common.legend = TRUE,
    legend = "bottom"
  ),
  nrow = 2,
  labels = "A"
)

fig_2_ABC
```

# 5 Modifications

The original study examined the average effects of financial incentives, freedoms, and vaccination access, but it did not consider participants' baseline willingness to vaccinate. This modification explores how baseline willingness interacts with other factors, particularly trust in government. The results show that trust has a stronger and more meaningful effect on increasing vaccination willingness than financial incentives, especially among hesitant individuals, highlighting the importance of accounting for heterogeneity in responses.

The first model examined baseline willingness and trust in government. The regression shows that, holding baseline willingness constant, a one standard deviation increase in trust in government is associated with a 0.0187 increase in the change in willingness to vaccinate (Estimate = 0.0187, $p < 2 \times 10^{-16}$). This demonstrates that individuals with higher trust in government are more likely to increase their willingness to vaccinate. Additionally, the interaction term between baseline willingness and trust in government is negative and significant (Estimate = $-0.0064$, $p = 2.43 \times 10^{-8}$). This means that for every one standard deviation increase in baseline hesitancy, the positive effect of trust in government on increasing willingness is slightly reduced by 0.0064.

| | Term | Estimate | Std.Error | t.value | p.value |
|---|---|---|---|---|---|
| 1 | (Intercept) | −0.001472356 | 0.001197223 | −1.229809 | 2.187827e−01 |
| 2 | baseline_std | −0.046454343 | 0.001313731 | −35.360624 | 6.465177e−266 |
| 3 | trust_gov_std | 0.018720479 | 0.001207654 | 15.501526 | 6.853514e−54 |
| 4 | baseline_std:trust_gov_std | −0.006411467 | 0.001148893 | −5.580561 | 2.427762e−08 |

Figure 4: Regression Results: Baseline Willingness × Trust in Government

In other words, trust in government especially helps individuals with initially low willingness to move toward vaccination. The intercept and baseline effects show that each one standard deviation increase in baseline willingness decreases the change in willingness by 0.0465 (Estimate $= -0.0465$, $p < 2 \times 10^{-16}$), consistent with the idea that those who are already willing have less room to increase. This analysis highlights a significant effect not included in the original study, which did not examine trust as a moderator of baseline willingness.

The second model investigated baseline willingness and Financial Incentive B to provide a comparison to the effect of trust in government. Holding baseline willingness constant, a one standard deviation increase in Financial Incentive B is associated with a 0.0111 increase in change in willingness (Estimate $= 0.0111$, $p < 2 \times 10^{-16}$). The interaction term with baseline willingness is marginally significant (Estimate $= -0.0021$, $p = 0.051$), indicating that for every one standard deviation increase in baseline willingness, the effect of the incentive slightly decreases. The main effect of baseline willingness remains negative and highly significant (Estimate $= -0.0358$, $p < 2 \times 10^{-16}$), showing that individuals who are already willing have less room to increase. Overall, while Financial Incentive B positively influences willingness, its effect is smaller and less robust than that of trust in government, highlighting that trust is more important than financial incentives in shaping changes in vaccination willingness.

|   | Term | Estimate | Std.Error | t.value | p.value |
|---|---|---|---|---|---|
| 1 | (Intercept) | −0.004132 | 0.001105 | −3.739 | 0.000185 |
| 2 | baseline_std | −0.035815 | 0.001105 | −32.413 | 0.000000 |
| 3 | financialB_std | 0.011145 | 0.001105 | 10.087 | 0.000000 |
| 4 | baseline_std:financialB_std | −0.002145 | 0.001098 | −1.954 | 0.050705 |

Figure 5: Regression Results: Baseline Willingness × Financial Incentive B

Policy Implications: Translating the coefficients into practical terms, these results suggest that trust in government has a stronger and more meaningful impact than financial incentives, particularly among initially hesitant individuals. For example, increasing trust in government by one standard deviation increases willingness by roughly 0.0187 units, whereas a comparable increase in financial incentives only raises willingness by 0.0111 units. Public health campaigns should therefore prioritize building and maintaining trust through transparent communication, engagement with trusted authorities, and responsiveness to public concerns. Financial incentives can complement these efforts but are less effective at motivating hesitant populations.

In summary, the statistical findings show that trust in government is both statistically significant and practically meaningful, while financial incentives have a positive but smaller effect. This highlights the importance of considering institutional trust in policy design for vaccination campaigns.

```
library(dplyr)
library(ggplot2)
library(ggeffects)
df_clean <- df %>%
  filter(!is.na(outcome_diff), !is.na(outcome_t1), !is.na(trust.
     government)) %>%
  mutate(
    baseline_std = as.numeric(scale(outcome_t1)),      # baseline
       willingness
    trust_gov_std = as.numeric(scale(trust.government))
  )
model_trust_gov <- lm(outcome_diff ~ baseline_std * trust_gov_std,
   data = df_clean)
summary(model_trust_gov)
coef_table <- data.frame(
  Term = c("(Intercept)", "baseline_std", "trust_gov_std", "baseline
```

```
    _std:trust_gov_std"),
  Estimate = c(-0.001472356, -0.046454343, 0.018720479,
    -0.006411467),
  Std.Error = c(0.001197223, 0.001313731, 0.001207654, 0.001148893),
  t.value = c(-1.229809, -35.360624, 15.501526, -5.580561),
  p.value = c(2.187827e-01, 6.465177e-266, 6.853514e-54, 2.427762e
    -08)
)
pdf("mod1_table.pdf", width = 8, height = 4)
grid.table(coef_table)
dev.off()
```

```
library(dplyr)
library(gridExtra)
library(grid)
df_clean <- df %>%
  filter(!is.na(outcome_diff), !is.na(outcome_t1), !is.na(financialB
    _diff)) %>%
  mutate(
    baseline_std = as.numeric(scale(outcome_t1)),
    financialB_std = as.numeric(scale(financialB_diff))
  )
model_finB <- lm(outcome_diff ~ baseline_std * financialB_std, data
   = df_clean)
summary(model_finB)
coef_table_finB <- data.frame(
  Term = c("(Intercept)", "baseline_std", "financialB_std", "
    baseline_std:financialB_std"),
  Estimate = c(-0.004132, -0.035815, 0.011145, -0.002145),
  Std.Error = c(0.001105, 0.001105, 0.001105, 0.001098),
  t.value = c(-3.739, -32.413, 10.087, -1.954),
  p.value = c(0.000185, 0, 0, 0.050705)
)
pdf("mod2_table.pdf", width = 8, height = 4)
grid.table(coef_table_finB)
dev.off()
```